

AMATH 732

Quiz 1; 30 Minutes

Question 1: 5 marks

i) Is it true that for $f(x) = x^2$ all Taylor polynomials are just the function itself? Explain.

ii) Consider $f(x) = xH(x)$ near $x = 0$. Since the derivative does not exist at $x = 0$ we cannot construct a Taylor series. How could we come up with a sensible linear approximation near $x = 0$?

Question 2: 5 marks

i) Consider the function

$$m(x) = \frac{1}{x} + \frac{\sin x}{x^2}$$

and its behaviour for large values of x . Without any formal theory explain what a useful approximation for the function is as $x \rightarrow +\infty$. Explain whether one should even try to find a power series approximation?

ii) Consider the function $g(x) = \sin^2(x)/x^2$. Define a new function $f(x)$ so that $g(x) = f(x)$ away from $x = 0$ and $f(x)$ is differentiable. How many non-zero terms will $P_{3,0}(x)$, the third order Taylor polynomial about $x = 0$ for $f(x)$, have. A sketch may be helpful in avoiding excessive work. The Taylor polynomial for sine is $\sin(m) \approx m - m^3/6 + m^5/120 + \dots$